WEB DEVELOPMENT LAB · WORKSHOP

From VS Code to GitHub Pages

Set up your tools, meet Copilot, and publish your own portfolio website

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TODAY’S PLAN

Four goals, one finished website

01

Set up VS Code

Install the editor, learn its layout and create your first file.

02

Connect to GitHub

Create an account, install Git and save your work online.

03

Turn on Copilot

Activate the free plan and learn to use an AI pair programmer responsibly.

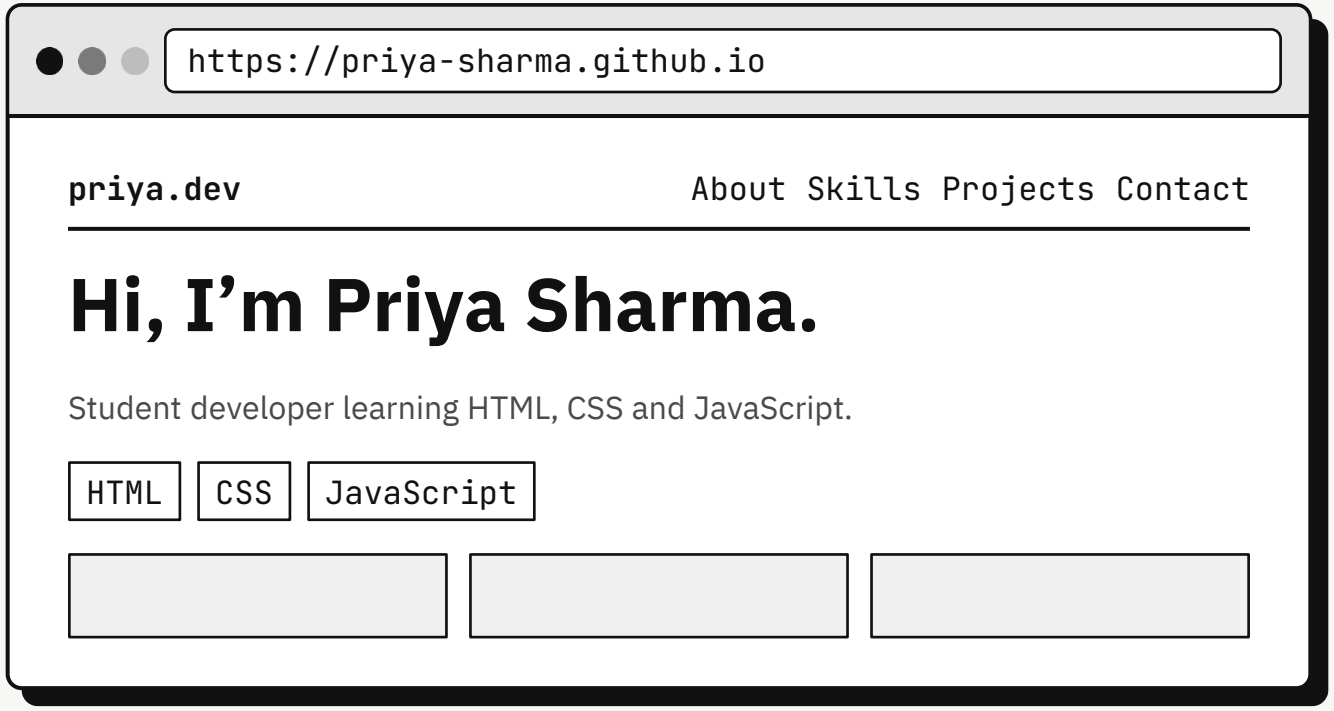
04

Build a portfolio

See two examples, then build and host your own page as the assignment.

WHERE WE ARE HEADING

Your own website, with your own address



By the end you will have

- A working setup: VS Code, Git, GitHub and Copilot
- A repository that stores every version of your code
- A public link you can share with recruiters

01

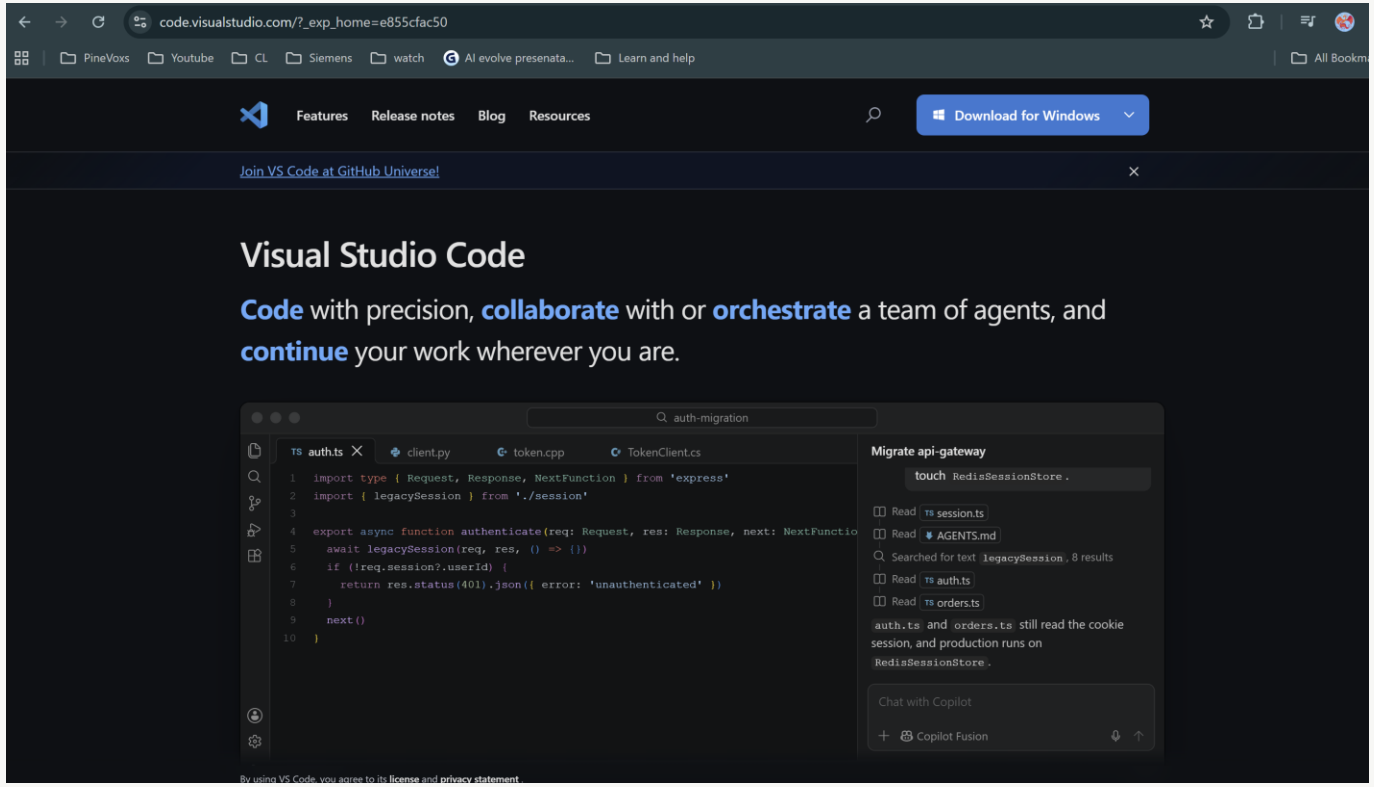
Setting up VS Code

The editor where you will write every line of your website

STEP 1 · INSTALL

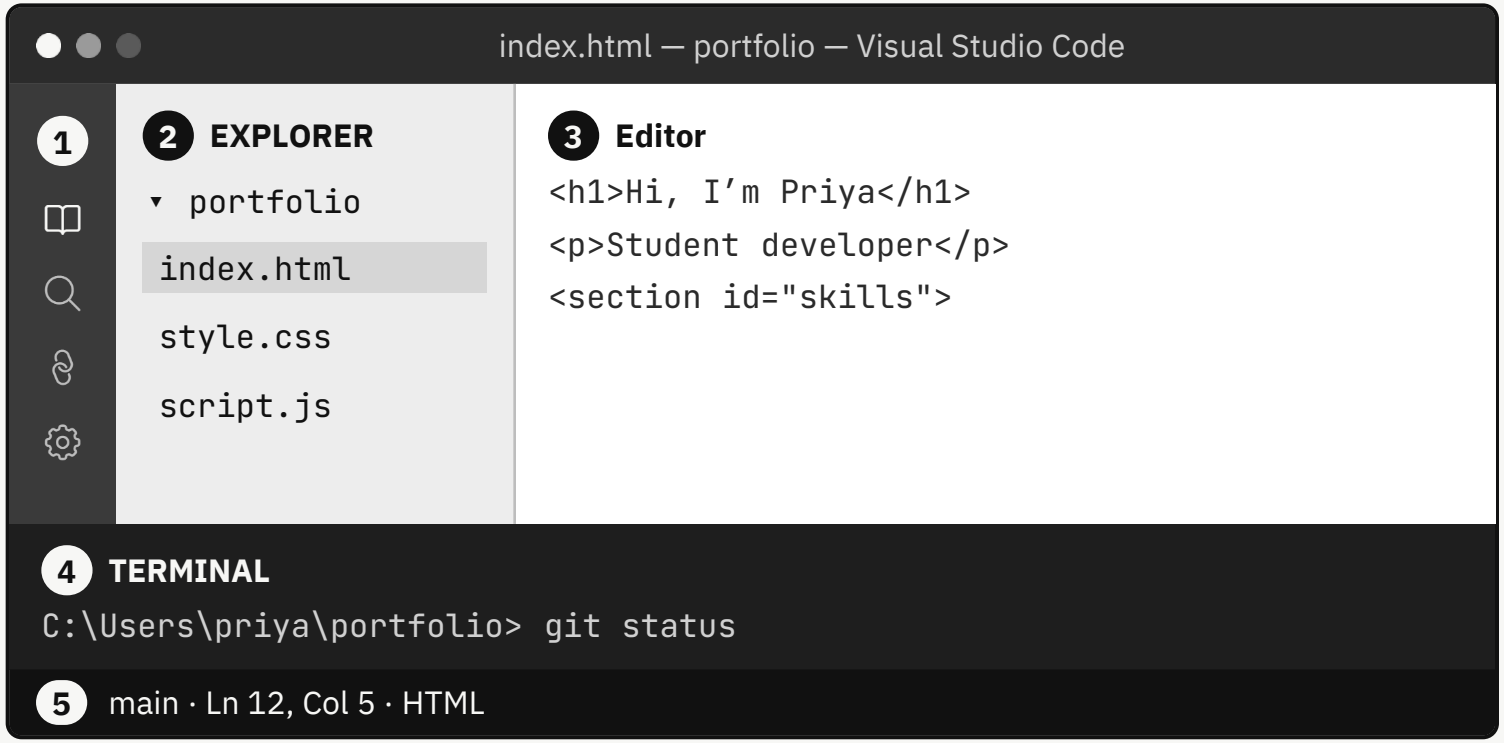
Download and install VS Code

1. Open **code.visualstudio.com** in your browser
2. Click the download button for your system (Windows, macOS or Linux)
3. Run the installer. On Windows, tick **Add to PATH** and **Add “Open with Code”**
4. Launch VS Code and pick a colour theme



STEP 2 · FIND YOUR WAY AROUND

Five areas of the VS Code window



1. **Activity Bar:** switch between Explorer, Search, Source Control and Extensions
2. **Side Bar:** your project's files and folders
3. **Editor:** where you type code
4. **Panel:** the Terminal, opened with Ctrl + `
5. **Status Bar:** branch, cursor position and Copilot status

STEP 3 · YOUR FIRST FILE

Create a project and preview it live

1. Make a folder called **portfolio**
2. **File › Open Folder** and choose it
3. New file: **index.html**
4. Type **!** and press **Enter** to get the HTML skeleton
5. In Extensions (Ctrl+Shift+X), install **Live Server**
6. Right-click the file › **Open with Live Server**

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Priya Sharma</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<h1>Hello, world!</h1>
<script src="script.js"></script>
</body>
</html>
```

02

Connecting to GitHub

Git keeps the history. GitHub keeps it online.

BEFORE WE CONNECT

Account, Git, and your identity

1. Create a GitHub account

Sign up at **github.com/signup**. Choose a professional username: it becomes your website address.

2. Install Git

Download from **git-scm.com** and keep the default options. macOS users can run **xcode-select --install**.

3. Tell Git who you are

Run the three commands on the right in the VS Code terminal.

TERMINAL PROBLEMS OUTPUT

```
$ git --version
git version 2.x.x
$ git config --global user.name "Priya Sharma"
$ git config --global user.email "priya@example.com"

Or use your GitHub noreply email to keep it private.
```

BEFORE YOU START

Stay safe: your site is public

OK

Fine to share

- Your first name or a display name
- Your skills, projects and learning goals
- A link to your GitHub profile
- A photo or email, only if you and a parent/guardian agree

STOP

Never publish

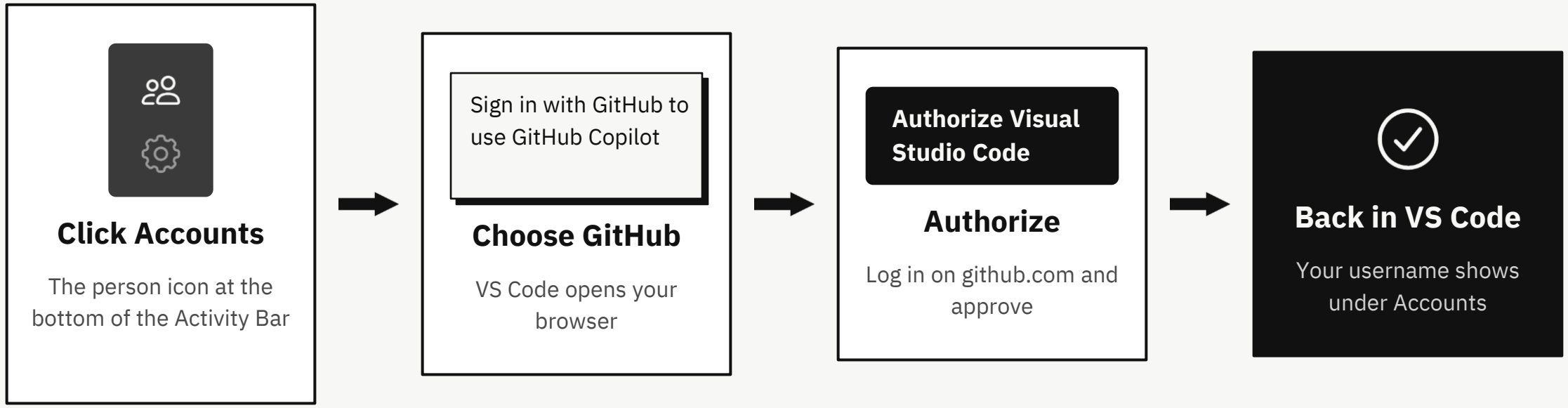
- Passwords, API keys or tokens
- Home address or phone number
- Your exact daily schedule
- Private information about anyone else

! GitHub requires you to be **at least 13**. If you are not, talk to your instructor and a parent/guardian about an approved alternative. Never sign up with a false age.

@ Want your email private? Use your **GitHub noreply address** in `git config user.email`. Check screenshots for private details too.

LINK THE TWO

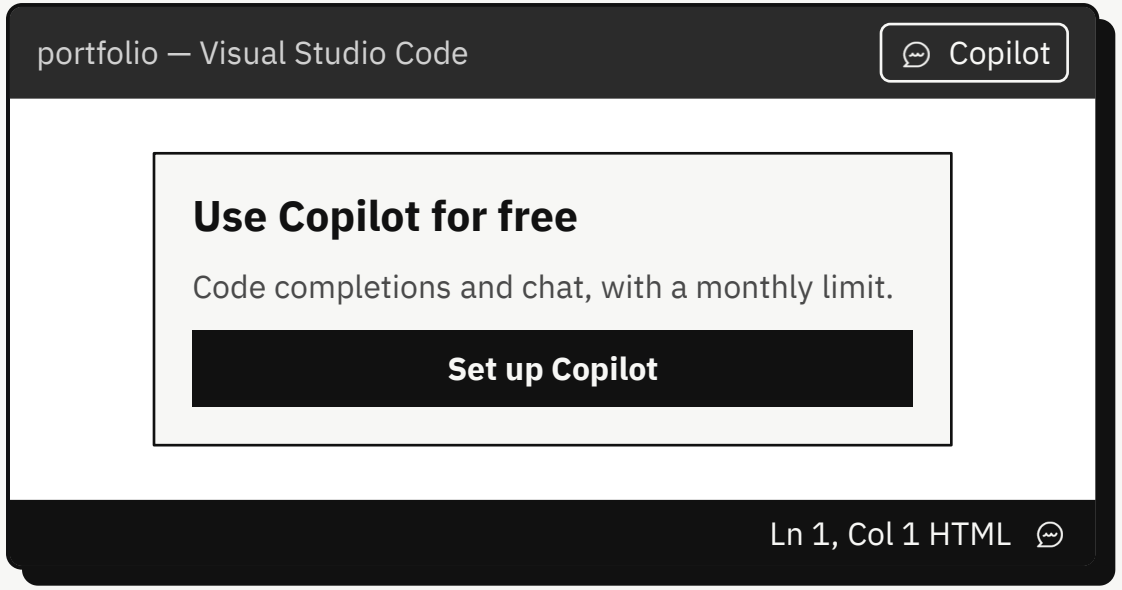
Sign in to GitHub from VS Code



TURN IT ON

Setting up Copilot in four clicks

1. Make sure you are signed in to GitHub in VS Code
2. Click the **Copilot icon** in the title bar or Status Bar
3. Choose **Set up Copilot**, then accept the free plan in the browser
4. Open Chat with **Ctrl+Alt+I** (Mac: Ctrl+Cmd+I)



SPEAK THE LANGUAGE

Six Git words you will use every day

Term	What it means	Think of it as
Repository	A project folder whose history Git tracks	A binder for one project
Commit	A saved snapshot, with a short message	A dated photo of your work
Branch	A line of work; ours is called main	A timeline of snapshots
Remote	The copy of the repository on GitHub	Your cloud backup
Push	Send your commits up to GitHub	Upload
Pull	Bring changes from GitHub down to you	Download

PART 1 · SET UP YOUR REPOSITORY

Create, clone, and add four files

1. Create a **public** repo on github.com

   Name it exactly `username.github.io`

2. Tick **Add a README file**, then create it

   This gives you something to clone.

3. In VS Code: **Ctrl+Shift+P › Git: Clone**

   Paste the repository URL and choose a local folder.

4. Open the cloned folder and create the files

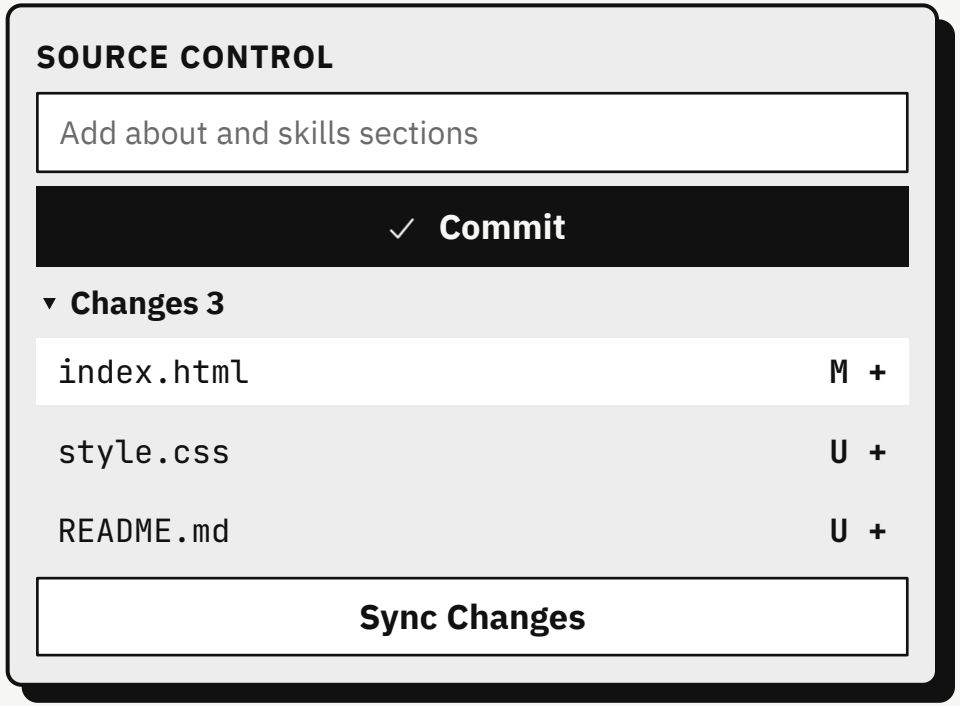
   Every file goes in the top-level folder.

File	Its job
`index.html`	Home page and all portfolio content
`style.css`	Colours, spacing, fonts and layout
`README.md`	Description, live link, how to preview locally
`learning_log.md`	Copilot log, test notes, reflection

Already have a `username.github.io` repository? Improve that portfolio instead of starting a new one.

THE EVERYDAY WORKFLOW

Save, commit and publish with Source Control



1. **Clone** your repo once: Ctrl+Shift+P › Git: Clone.
2. **Stage** files with **+**. M means modified, U means new.
3. Write a clear message and click **Commit**.
4. **Sync Changes** (push) after each commit; **Pull** when you start.

PART 1 · HABITS THAT EARN POINTS

Three meaningful commits, and one pull

Your commit history should tell the story

`Create portfolio structure`	First commit
`Add project showcase and styling`	Real progress
`Fix navigation and improve mobile layout`	After first deploy

Avoid messages like `update` or `stuff`. Each commit should show visible progress.

Pull at the start of each session

- Open the Source Control view, then **… › Pull** (or Sync Changes).
- **Pull** downloads changes from GitHub. **Push** uploads your commits.

Evidence you will submit

- A screenshot of the project open in VS Code
- A link to the repository's commit history

03

GitHub Copilot, free

An AI pair programmer inside your editor

TWO WAYS TO WORK WITH IT

Inline suggestions and chat

As you type

```
<!-- three project cards -->
<div class="projects">
<article class="card">
<h3>Calculator</h3>
</article>
```

Tab accepts the grey text. **Esc** dismisses it.

Copilot Chat

Explain what display: flex does in my style.css

Flexbox lays out the children of an element in a row or column…

Why is my navigation not centred?

Ask it to **explain** and **fix**, not just to write.

⚠ **Rule of the class:** never accept code you cannot explain. You will be asked about your own page.

PART 3 · YOUR COPILOT TASKS

Three purposeful Copilot interactions

Type	What to ask for	Example prompt you can adapt
1 Build	Help creating or improving one small part of the site	Help me create a beginner-friendly HTML structure for a student portfolio with About Me, Skills and Projects. Use placeholders instead of inventing facts about me.
2 Understand	An explanation of some HTML or CSS	Explain how the navigation links connect to section IDs in my HTML.
3 Improve / fix	Fix a bug, improve readability, or make it work on mobile	My project cards overflow on a narrow screen. Help me find the CSS causing this and explain a simple fix.

! Use **Copilot inside VS Code**, and record all three in `learning_log.md` (next slide).

PART 3 · DOCUMENT YOUR LEARNING

Keep a learning log

Prompt or question	What Copilot suggested	Accepted, changed or rejected?	How I checked it
Your exact prompt	A short summary	Your decision, and why	A test or observation

Reflection: 150–250 words

- How do GitHub, VS Code, Copilot and GitHub Pages each help you build?
- One challenge, and how you solved it
- One piece of code you can now explain
- What you would improve next

AI ground rules

- Read every suggestion, personalise it, test it
- Be ready to explain your files and make a small edit live
- No unchanged AI site; no copying a classmate
- Credit templates, tutorials and borrowed assets in README

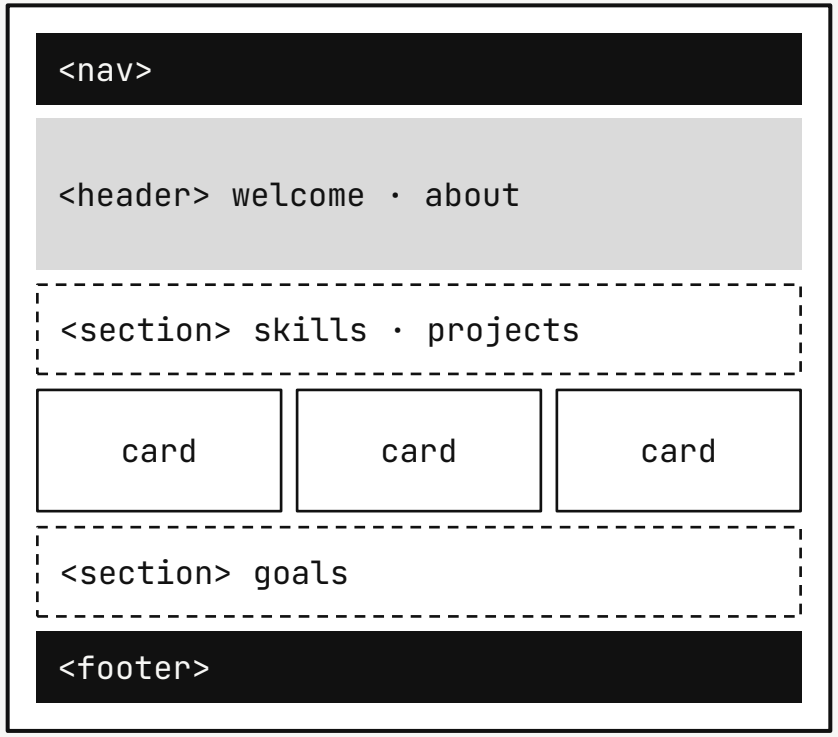
04

Your portfolio page

HTML for structure, CSS for style, JavaScript for behaviour

PLAN BEFORE YOU CODE

The anatomy of a portfolio page



Section	What goes in it
Navigation	Links that jump to each section
Welcome + About	Display name, headline, a short paragraph about you
Skills	At least four skills or tools, described honestly
Projects	At least two; one Python or data science
Learning goals	What you'll learn next, or how you'll help others
Footer	Link to your GitHub profile (email optional)

PART 2 · PROJECTS

Every project entry needs four things

01

Title and purpose

What problem does it solve, or why did you build it?

02

Tools used

For example Python, pandas, matplotlib, HTML, CSS

03

What you learned

Or what you contributed, in your own words

04

A working link

Code, notebook, demo or other artifact the instructor can open

! At least **two** projects; at least **one Python or data science** project. Your portfolio itself can be the second.

i GitHub Pages only hosts static files and **cannot run Python**. Link to your notebook or repository. Private repo? Give your instructor access.

THREE FILES, THREE JOBS

How the page is put together

index.html

Structure

What is on the page

```
<section id="skills">
<h2>Skills</h2>
<li>HTML</li>
</section>
```

style.css

Style

How it looks

```
.projects {
display: grid;
gap: 16px;
}
```

script.js

Behaviour

Optional: what it does when you click

```
btn.addEventListener(
"click", () => {
alert("Hello!");
});
```

PART 2 · DESIGN AND USABILITY

Make it readable for everyone

- ✓ **Navigation that works** · Links like `#projects` jump to matching section ids
- ✓ **Desktop and mobile** · Readable at a narrow, phone-sized width
- ✓ **Meaningful link text** · “My data project”, not “click here”
- ✓ **No placeholders** · Remove sample text and check spelling
- ✓ **Your own style.css** · Linked from index.html and personalised
- ✓ **Clear and readable** · Headings, font size and strong text contrast
- ✓ **Images done right** · Alt text, and only images you may use
- ✓ **GitHub profile link** · A working link to github.com/username

Scope: plain HTML and CSS are enough. JavaScript is optional. No custom domain, database or contact form needed.

Examples

1. https://anushka05012.github.io/
2. https://eshaandhavala.github.io/

GO LIVE

Host it free with GitHub Pages

1. Create a **public** repository named exactly **username.github.io**
2. Push your files. **index.html** must sit at the top level
3. Open **Settings › Pages**
4. Source: **Deploy from a branch**. Branch: **main**, folder **/ (root)**. Save
5. Visit **https://username.github.io**. Allow up to 10 minutes

1. Two types of GitHub Pages repositories

Suppose your GitHub username is `anushka123`.

Feature	Personal website	Project website
Repository name	`anushka123.github.io`	`fitness-app`
Website URL	`anushka123.github.io`	`anushka123.github.io/fitness-app/`
Purpose	Showcase yourself	Showcase one specific project
Example content	Resume, skills, projects, experience	Fitness application demo
Number per account	One primary repository of this type	Multiple project repositories

PART 4 · TEST

Prove the update, then test everything

After the first deploy

1. Make one improvement in VS Code
2. Commit and push it
3. Confirm the live site changed

This can be one of your three commits. Deployment failing? Check the repository's **Actions** tab.

Test checklist · record results in `learning_log.md`

- [ ] Live URL opens in a private/incognito window
- [ ] All required sections hold your own content
- [ ] Navigation, GitHub profile and project links work
- [ ] Readable on desktop and at a narrow width
- [ ] Images (if any) load and have alt text
- [ ] Your latest change shows on the live site
- [ ] No private info on the site, repo or screenshots

WHEN SOMETHING GOES WRONG

Common problems and quick fixes

You see	Try this
404 on your github.io link	Check repo name, Pages branch/folder, index.html at top level, Actions tab
Page has no styling	Check href="style.css": names are case-sensitive online
Changes not showing	Did you push? Wait a few minutes, then refresh with Ctrl+F5
git is not recognized	Install Git, then restart VS Code
Push rejected	Pull or Sync first, then push again
Copilot does not respond	Check sign-in and monthly limit; still blocked? Tell your instructor

PART 4 · SUBMIT

What to hand in

1. **Live portfolio URL**

   `https://username.github.io/`

2. **Source repository URL**

   `github.com/username/username.github.io`

3. **Commit history URL**

   `Commits page of your repository`

4. **Learning log link**

   `learning_log.md in your repository`

5. **Two screenshots**

   `Project open in VS Code · live site in a browser`

README.md must include

- A short project description
- The live website link
- How to preview locally (open index.html)
- Credits for templates or borrowed assets

Improving an existing portfolio? Summarise what is new in the learning log and name the assignment commits. Submit through your instructor’s submission channel.

Questions?

Then go build something that is yours.

Editor

`code.visualstudio.com`

Version control

`git-scm.com`

Hosting guide

`docs.github.com/pages`

Student benefits

`education.github.com`